Journal of Medical Virology

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JOURNAL OF MEDICAL VIROLOGY

SHORT COMMUNICATION OPEN ACCESS

Association Between NK Cell Genetic Variants and the Development of Long COVID Associated- and Prepandemic Small Fiber Neuropathy

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Received: 25 July 2024 | **Revised:** 18 October 2024 | **Accepted:** 11 November 2024

Funding: The study was supported by Medical and Scientific Fund of the Mayor of the Federal Capital of Vienna, Austria and Center for Virology, Medical University of Vienna, Austria.

Keywords: long COVID | NK cell | NKG2C | small fiber neuropathy

ABSTRACT

Long coronavirus disease 2019 (COVID) (LC) symptoms including pain and autonomic dysfunction are in some patients associated with small-fiber neuropathy (SFN). The pathomechanisms underlying SFN are mostly unclear. Natural killer (NK) cells play a crucial role in immune regulation, viral clearance and nerve metabolism. The aim of this study was to identify associations between development of small-fiber dysfunction dependent and independent of severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) infection, and human genetic markers associated with specific NK cell functions. The genetic markers assessed in all cohorts included: *FCGR3A*, *IGHG1*, *HLA-E*, *NKG2C*, and rs9916629. Genotyping was performed using TaqMan assays, Sanger sequencing and touchdown polymerase chain reaction. We assessed human cytomegalovirus (HCMV) IgG serostatus in all participants, and screened for anti-neuronal, anti-glial and anti-ganglioside autoantibodies in both patient cohorts. We included 50 LC patients with newly-emerged symptoms of small-fiber dysfunction after SARS-CoV-2 infection, 27 prepandemic SFN patients and 320 control persons. Markers associated with low NKG2C response, that is, deletion of the *NKG2C* gene and lack of prior HCMV infection (IgG seronegativity), occurred significantly more frequently in prepandemic SFN patients compared to LC patients and controls ($p = 0.0109$ and 0.0005, respectively). In conclusion, markers of impaired NKG2C pathways are associated with prepandemic SFN, but not with Long COVID-associated small-fiber dysfunction.

1 | Introduction

Long coronavirus disease 2019 (COVID) (LC) represents a heterogeneous complex of symptoms and is often characterized by systemic symptoms like chronic pain, extensive fatigue, and postexertional malaise. These symptoms may partly result from small-fiber dysfunction, which is increasingly diagnosed in LC patients as small-fiber neuropathy (SFN) [1, 2].

SFN defines a clinical picture resulting from damage to thinly-myelinated Aδ-fibers or unmyelinated C-fibers, which are involved in the perception of pain and temperature, and in the regulation of autonomic body functions [3]. Following small–fiber damage, patients experience paresthesia and neuropathic pain, and can develop autonomic dysfunctions including exertional intolerance or orthostatic dysregulation. SFN development has been observed in the scope of various conditions

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including infectious or immune-mediated diseases, where nerve damage is often supposed to result from immune dysregulation or autoimmune processes [3]. In many cases, however, the pathogenesis underlying small-fiber damage remains unclear.

Natural killer (NK) cells play a pivotal role in the prevention of autoimmune processes and in the control of viral infections based on their cytotoxicity and ability of cytokine release [4]. They are also involved in nerve homeostasis, as they selectively destroy damaged nerves and thus may prevent sensitization and chronic pain development [5]. NK cell functions are regulated by inhibitory and activating signals. Activating pathways include the FcγRIIIA–IgG and the NKG2C/CD94–HLA-E axis, both of which are functionally determined by human genetic variants influencing the efficacy of NK cell activation in the individual host.

The FcγRIIIA–IgG axis mediates antibody-dependent cellular cytotoxicity (ADCC). Its function is determined by a genetic dimorphism in *FCGR3A* (F158V), encoding the NK cell FcγR-surface receptor FcγRIIIA (CD16A) which mediates activating signals when binding to IgG1. Human IgG1 occurs as different allotypes based on a polymorphism in the *IGHG1* region, termed gamma 1 marker (G1m) 3/17. Both FcγRIIIA and G1m variants differentially affect antibody-receptor binding affinity and the execution of ADCC functions [6].

NKG2C is another activating NK cell surface receptor, which binds to the MHC molecule HLA-E expressed by stressed or virus-infected cells [7]. NKG2C is highly expressed on "adaptive" NK cells, which expand in response to human cytomegalovirus (HCMV) infection. Deletion of the *NKG2C* gene is observed in a part of the population and has been shown to reduce NKG2C+ cell functions, with implications for various infectious and autoimmune diseases [8–10]. Binding efficacy to NKG2C is also dependent on the HLA-E variant occurring in the individual host, as determined by its genetic variants *HLA-E*0101* or **0103*, which differently influence the presentation of virus peptides by HLA-E and the extent of NK cell stimulation [7].

Finally, a polymorphism at position rs9916629 (C/T) has been shown to impact the composition of NK cell subpopulations within the host, individually altering NK cell functions [11].

As NK cells are involved in nerve metabolism as well as in immune regulation and virus control, we investigated whether there is an association between genetic predispositions influencing NK cell functions and small-fiber dysfunction in LC patients and in patients with idiopathic or immune-mediated SFN not associated with prior severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) infection.

2 | Methods

2.1 | Study Cohorts

In this study, 50 LC patients were analyzed. Inclusion criteria were newly reported neuropathic symptoms characteristic of small-fiber dysfunction as shown in Supporting Information: Table S1, emerging within 3 months after SARS-CoV-2 infection as confirmed by polymerase chain reaction (PCR) ($n = 47$, 94%) or antigen test ($n = 2$, 4%) from nasopharyngeal swab, or by SARS-CoV-2 IgG seroconversion when COVID-19 vaccination was not yet available ($n = 1$, 2%). In 15 LC patients (30%), SFN was confirmed by reduced intraepidermal nerve fiber density (IENFD) in skin biopsy as described previously [12] and evaluated accordingly [12, 13]. LC patients were included either prospectively ($n = 30$, 60%) during a study visit at the Center for Virology, or retrospectively ($n = 20$, 40%) using sera stored in the biobanks of the Center for Virology and the Division of Neuropathology and Neurochemistry, Department of Neurology, Medical University of Vienna.

Additionally, we included a cohort of 27 SFN patients with symptom onset before 02/2020 ("prepandemic SFN"). SFN was diagnosed in all cases by the presence of neuropathic SFN symptoms plus reduced IENFD in skin biopsy as described above. Prepandemic SFN patients were included either prospectively ($n = 2$, 7%), or retrospectively ($n = 25$, 93%) as described above.

Clinical information of retrospectively included patients was obtained by medical records.

As a population control, we included 320 control persons matched for age and sex with LC and prepandemic SFN patients, whose plasma samples were initially sent to the Center for Virology for routine vaccination titer control.

2.2 | Genotyping

Genotyping was performed in serum or plasma samples. Genomic DNA was isolated from serum or plasma using the NucliSens easyMAG extractor (BioMérieux, France) and eluted in 50 μL H_2O. The following genetic markers influencing NK cell functions were assessed in all cohorts: *FCGR3A* (FcγRIIIA V/F), *IGHG1* (G1m 3/17), *HLA-E* (HLA-E*0101/*0103), *NKG2C* (NKG2C wt/del) and rs9916629 (C/T). Genotyping was performed by TaqMan assays (*FCGR3A*, *HLA-E*, rs9916629), Sanger sequencing (G1m) and touchdown PCR (*NKG2C*) as previously described [11, 14–16].

2.3 | HCMV Serostatus Assessment

Human cytomegalovirus (HCMV) IgG serostatus was assessed by a commercial ELISA (Anti-CMV ELISA (IgG), Euroimmun, Germany).

2.4 | Autoantibody Testing

A subset of LC and prepandemic SFN patients, of whom enough leftover material was available, were screened for a panel of serum autoantibodies by commercially available and in-house assays (Supporting Information: Table S2).

2.5 | Ethical Considerations

This study was approved by the Ethical Committee of the Medical University of Vienna (EK-no. 1815/2022, biobank: EK-no. 1926/2020, EK-no. 1123-2015). Written informed consent was obtained by prospectively included patients before study participation.

2.6 | Statistics

Demographic data and genotype distributions were compared by χ^2 test and Kruskal–Wallis test where applicable. The significance level α was set at 0.05. All statistics were performed with GraphPad Prism 9.2.0.

3 | Results

3.1 | Study Cohorts

To examine the association between genetic NK cell markers and small-fiber dysfunction dependent and independent of SARS-CoV-2, we investigated 50 LC patients with newly emerged clinical signs of small-fiber dysfunction after SARS-CoV-2 infection, 27 prepandemic SFN patients, and 320 age- and sex-matched control persons. Demographic details of the cohorts are summarized in Table 1. Neuropathic symptoms reported are listed in Supporting Information: Figure S1. Details on symptom development in LC patients are depicted in Supporting Information: Table S3.

3.2 | Genetic NK Cell Markers Associated with ADCC, and rs9916629

To reveal whether genetic markers influencing NK cell-mediated ADCC or rs9916629 genotypes are associated with small-fiber dysfunction, we compared genotype distributions of *FCGR3A* (V/F), G1m (3/17) and rs9916629 (C/T) between the study cohorts. We observed no difference in the respective genotype distributions between LC patients, prepandemic SFN patients and controls (Figure 1A–C).

3.3 | Genetic NK Cell Markers Associated With the NKG2C/CD94–HLA-E Axis

To analyze the association between NK cell markers affecting the NKG2C/CD94–HLA-E pathway and small-fiber dysfunction, we assessed the *HLA-E* and *NKG2C* genotypes in patients and controls. *HLA-E* genotype distribution was similar between all cohorts (Figure 2A). However, homozygous deletions of *NKG2C* occurred significantly more frequently in prepandemic SFN patients compared to the population controls ($p = < 0.0001$) (Figure 2B). As NKG2C receptor expression and expansion of NKG2C+ cells are significantly promoted by HCMV infection, HCMV IgG serostatus was assessed in all participants. HCMV IgG seropositivity rates did not differ significantly between LC patients (50%), prepandemic SFN patients (35%) and the population controls (55%).

We then grouped patients and controls according to their NKG2C response profile. A high NKG2C response profile was

TABLE 1 | Demographic data of study cohorts.

	Long COVID *n* (%)	Prepandemic SFN *n* (%)	Population control *n* (%)	*p*-Value
Total	50 (100)	27 (100)	320 (100)	
Female sex	39 (78)	14 (52)	222 (69)	ns
Median age in years [range]	39 [17–76][a]	38 [13–70][a]	39 [18–76]	ns
Diagnosis of small-fiber dysfunction				
Skin-biopsy confirmed SFN	15 (30)	27 (100)		
Clinical diagnosis	50 (100)	27 (100)		
SFN type in skin-biopsy confirmed cases				
Length-dependent	5 (33)	17 (63)		
Non-length-dependent	10 (67)	10 (37)		
Cause of small-fiber dysfunction				
Post COVID	50 (100)	0 (0)		
Idiopathic	0 (0)	22 (81)		
Immune-mediated[b]	0 (0)	4 (15)		
Vasculitic neuropathy	0 (0)	1 (4)		
Type of symptoms reported				
Neuropathic	50 (100)	27 (100)		
Autonomic	44 (88)	19 (70)		

Abbreviations: COVID, coronavirus disease 2019; ns, not significant; SFN, small fiber neuropathy.
[a]At symptom onset.
[b]SFN was defined as immune-mediated when it occurred in the scope of underlying autoimmune diseases.

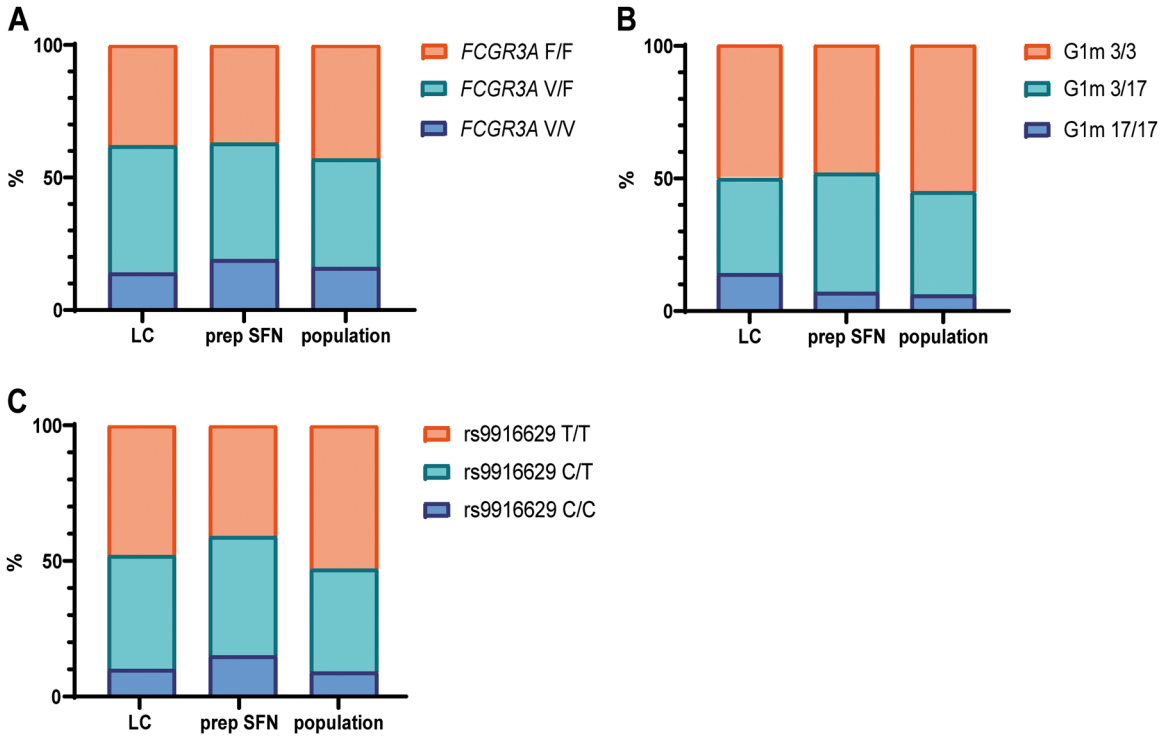

FIGURE 1 | Genetic natural killer (NK) cell markers associated with antibody-dependent cellular cytotoxicity (ADCC) and the rs9916629 polymorphism. Genotype distributions of (A) *FCGR3A*, (B) G1m and (C) rs9916629 in Long COVID patients (LC), prepandemic small fiber neuropathy (SFN) patients (prep SFN) and population controls (population). Comparisons of genotype distributions were performed by χ^2 test.

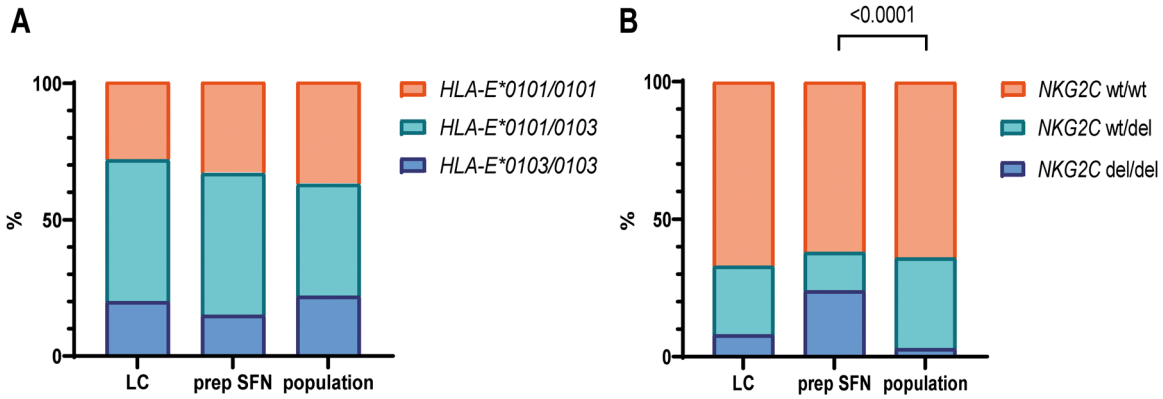

FIGURE 2 | Genetic natural killer (NK) cell markers associated with the NKG2C/CD94–HLA-E axis. Genotype distributions of (A) *HLA-E* and (B) *NKG2C* in Long COVID patients (LC), prepandemic small fiber neuropathy (SFN) patients (prep SFN) and population controls (population). For *NKG2C* genotyping, leftover material was available from 48/50 LC patients and from 21/27 prepandemic SFN patients. Brackets show statistically significant differences in genotype distribution as assessed by χ^2 test.

defined as HCMV IgG seropositivity and presence of the *NKG2C* wt (wt/wt or wt/del) allele, while a low NKG2C response profile was characterized by either HCMV IgG seronegativity, or by status of HCMV IgG seropositivity with presence of the *NKG2C* del/del genotype. As shown in Figure 3A, the proportion of patients with low NKG2C responses was significantly higher in prepandemic SFN patients (84%) compared to LC patients and population controls (52% and 47%; $p = 0.0109$ and 0.0005, respectively). This observation was confirmed when prepandemic SFN patients were compared to a subgroup of 15 LC patients in whom SFN was diagnosed by skin biopsy ($p = 0.0024$) (Figure 3B).

3.4 | Autoantibody Assessment

To assess the role of autoimmune mechanisms in small-fiber dysfunction, we investigated a panel of well-characterized autoantibodies in 46 LC and 24 prepandemic SFN patients (Supporting Information: Table S2). Anti-neuronal and anti-glial antibodies against intracellular and surface proteins were negative in all LC cases. Two/46 LC samples were weakly positive for anti-GM1 IgM ganglioside antibodies. One/24 prepandemic SFN patients had low titers of glycine receptor antibodies, and another had low titers of anti-GD1a IgG ganglioside antibodies.

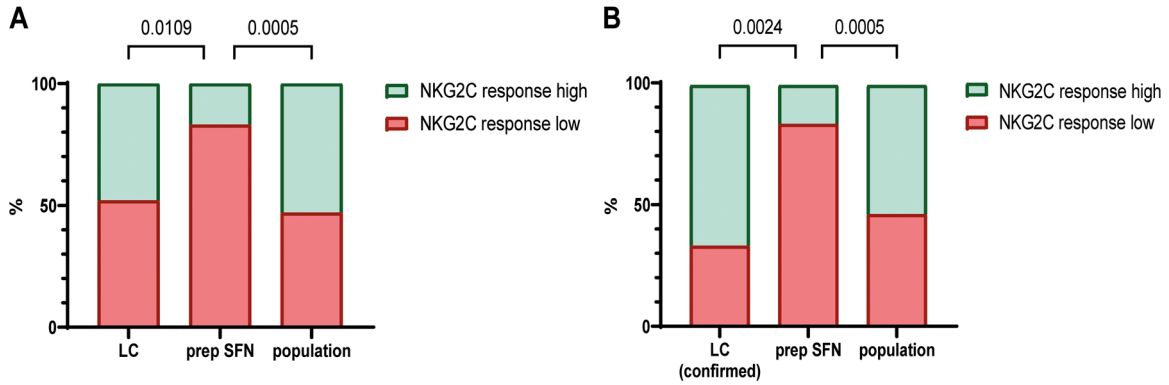

FIGURE 3 | Prevalence of patients with NKG2C high and low responses in the study cohorts. "NKG2C response high" indicates patients who are HCMV IgG seropositive and carry the *NKG2C* wt allele (wt/wt or wt/del). "NKG2C response low" indicates patients who are HCMV IgG seronegative, or who are HCMV IgG seropositive while displaying the *NKG2C* del/del genotype. Comparison between (A) Long COVID patients (LC) or (B) LC patients with skin biopsy confirmed small fiber neuropathy (SFN) (LC (confirmed)), respectively, and prepandemic SFN patients (prep SFN) as well as population controls (population). Brackets show statistically significant differences in genotype distribution as assessed by χ^2 test.

4 | Discussion

In the present study, we show that patients with SFN unrelated to SARS-CoV-2 infection exhibited markers of impaired NKG2C functions significantly more frequently than healthy controls, while this was not observed for patients with LC-associated signs of small-fiber dysfunction, including patients with confirmed LC-SFN.

The loss of NKG2C due to the homozygous deletion of the *NKG2C* gene has previously been associated with impaired NK cell immunity, negatively impacting virus control and the killing of autoreactive cells [9, 10]. We identified the *NKG2C* del/del genotype in only 3% and 8% of the control population and LC patients, respectively, which is in agreement with earlier data [8, 17]. In contrast, we observed the *NKG2C* del/del genotype in 24% of the prepandemic SFN patients. When taking into account the individual HCMV infection status, a total of 84% of the prepandemic SFN patients showed markers of low-level NKG2C responses, either due to HCMV IgG seronegativity and/or the presence of the *NKG2C* del/del genotype. This proportion was significantly higher than that observed in LC patients (52%) and controls (47%). Thus, our data suggest that impaired NK cell responses based on the NKG2C pathway could contribute to the development of idiopathic or immune-mediated SFN unrelated to SARS-CoV-2 infection. Interestingly, this mechanism is unlikely to represent one of the primary causes of LC-associated small-fiber dysfunction.

It is unclear why NKG2C-mediated pathways differently affect small-fiber functions in prepandemic SFN versus LC patients. SFN development has been described in the scope of human immunodeficiency virus or hepatitis C virus infection, and in our study, several prepandemic idiopathic SFN patients reported SFN onset shortly after a viral infection, most commonly with Epstein-Barr virus or influenza virus [18, 19]. Functional impairment of the NKG2C pathway might be associated with a dysregulation of immune responses after these viral infections, including reduced regulation of autoimmune processes, which might contribute to small-fiber damage. On the other hand, NK cells are involved in the selective destruction of nerves pre-damaged by different causes, and SFN could result from ineffective clearance of such pre-damaged nerves via the NKG2C pathway [5]. Recent evidence indicates that small-fiber damage in LC could result from cytokine-mediated sensitization of nociceptive nerve fibers, a pathomechanism potentially unaffected by NK cell-mediated control of acute SARS-CoV-2 infection or targeted nerve clearance [1].

We did not observe any association between SFN development and genetic variants influencing ADCC or the rs9916629 polymorphism. We did not find clinically relevant levels of autoantibodies in either patient cohort. The low anti-GM1 IgM titers found in two LC patients are supposedly part of the natural antibody repertoire, as these antibodies have the potential to induce autoimmune neuropathy only if they show increased affinity, high abundance, and a class switch to IgG [20]. Thus, nerve-targeting autoimmune processes involving the autoantibodies tested here do not seem to play a role in LC-associated small-fiber dysfunction. Similarly, the low titers of glycine receptor and GD1a antibodies in prepandemic SFN patients are most likely irrelevant.

In our LC cohort, the diagnosis of small-fiber dysfunction was mainly based on clinical symptoms, and large-fiber involvement was not excluded in all patients, which pose limitations to the study. However, we could confirm our main finding with a subgroup of LC patients with skin biopsy confirmed SFN ($n = 15$, 30%), validating the results of our overall analysis.

The results of this study reveal differences in the pathogenesis of small-fiber dysfunction developing in the scope of Long COVID versus independently of SARS-CoV-2 infection. Importantly, idiopathic or immune-mediated SFN independent of LC seems to be associated, to a certain extent, with impaired NKC2C+ cell functions. Further studies will be needed to confirm these data and to develop therapeutic interventions tailored to different types of SFN.

Author Contributions

Marianne Graninger and **Elisabeth Puchhammer-Stöckl:** conceptualization. **Marianne Graninger** and **Elisabeth Puchhammer-**

Stöckl: writing. **Verena Endmayr**, **Paulus Rommer**, **Sigrid Klotz**, **Ellen Gelpi**, **Hannes Vietzen**, and **Romana Höftberger:** editing. **Marianne Graninger**, **Verena Endmayr**, **Laura M. Kühner**, **Sarah M. Berger**, **Paulus Rommer**, **Sigrid Klotz**, **Ellen Gelpi**, **Hannes Vietzen**, and **Romana Höftberger:** methodology and investigation. **Marianne Graninger** and **Elisabeth Puchhammer-Stöckl:** resources, funding acquisition and supervision. All authors contributed to the article and approved the submitted version.

Acknowledgments

We thank Thomas Urbanek, Gabrielle Sigmund, Claudia Kellner, Michaela Binder, Barbara Dalmatiner, Johanna Wachutka, Katrin Köbl, Viktoria Stark, Ines Lukac, Judith Ludwig, Andreas Bugelnig, Benjamin Koch, and Silvia Zimmermann for excellent technical assistance. This study was funded by the Medical and Scientific Fund of the Mayor of the Federal Capital of Vienna, Austria, and by the Center for Virology, Medical University of Vienna.

Ethics Statement

This study was approved by the Ethical Committee of the Medical University of Vienna (EK-no. 1815/2022, biobank: EK-no. 1926/2020, EK-no. 1123-2015).

Consent

Written informed consent was obtained by prospectively included patients before study participation.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data obtained in this study is treated as confidential and is not freely available. Research data are not shared.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.